✅ STUDENT MODULE (USER SIDE FEATURES)

🔐 1. Authentication & Account Management

- Student Registration
- Student Login
- JWT-based Authentication
- Password Encryption
- Logout
- Account Deletion
- Session Management

👤 2. Student Profile Management

- Enter Personal Details
  - Full Name
  - Email
  - Mobile Number
- Academic Details
  - Current Level (10th / 12th)
  - Stream (MPC, BiPC, MEC, CEC, HEC, Others)
  - Subject-wise Marks
- Location Details
  - State
  - District
- Automatic Aggregate Calculation
- View Profile
- Update Profile

🧪 3. Aptitude & Interest Assessment

- Online MCQ-Based Test
- Categories:
  - Logical Ability

    - Numerical Ability
    - Verbal Ability
    - Creativity
    - Interest Areas
- 15–20 Questions
- Auto Score Calculation
- Multiple Attempts Allowed
- Latest Attempt Considered
- Retake Assessment Option

🎯 **4. Career Recommendation Engine**

- Career Suggestions Based On:
    - Academic Marks
    - Stream
    - Aptitude Scores
- Top 3 Career Recommendations
- Confidence Percentage for Each Career
- Rank-Based Ordering
- Domain Filtering (cse,csm,aiml,civil etc. I specified below)
- Explanation of Recommendation
- Hybrid Logic (ML + Rule-Based)

🏫 **6. Course & College Mapping**

- Career → Course Mapping
- Course → College Mapping
- Government Colleges First Priority
- Filter By:
    - Stream
    - District
    - State

- Display:
    - College Name
    - Location
    - Clg official link

🗺️ **7. Career Roadmap Visualization**

- Step-by-Step Flow:
    - Career → Degree → Entrance Exam → Job Role
- Required Skills
- Higher Studies Options
- Career Growth Levels
- Salary Range (Basic)
- Visual Tree / Flowchart Display

📊 **8. Career Outcome Visualization**

- Growth Path
- Demand Level
- Average Salary
- Future Scope
- Career Comparison

📚 **9. Learning Resource Hub**

- Open Source E-Books
- Free Course Links
- Video Tutorials
- Skill Roadmaps
- Career-Based Resource Filtering
- Search Resources
- Download PDFs

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10. Expert Chat System

- View Available Experts
- Expert Specialization Display
- Start Chat with Expert
- Send Questions
- Receive Replies
- Chat History

🤖 11. AI Chatbot Advisor

- Ask Career Questions
- Ask Course Suggestions
- Ask Skill Guidance
- Profile-Aware Responses
- FAQ Handling

🔄 12. Progress Tracking

- Stores Assessment History
- Stores Recommendation History
- Shows Last Recommendation
- Shows Last Test Date

🔔 13. Notifications

- Profile Completion Reminder
- Assessment Reminder
- New Resource Notification
- Scholarship Updates

⭐ 14. Feedback System

- Rate Recommendation
- Give Suggestions
- Report Issue

✅ ADMIN MODULE (MANAGEMENT SIDE)

🔐 1. Admin Authentication

- Admin Login
- Secure Role-Based Access

📊 2. Admin Dashboard

- Total Students
- Total Assessments
- Popular Careers
- Active Users

🧠 3. Career Domain Management

- Add Domain
- Edit Domain
- Delete Domain

🎯 4. Career Role Management

- Add Career Role
- Assign Domain
- Edit Role
- Delete Role

🏫 5. College Management

- Add College
- Update College
- Delete College
- Manage:
    - Location
    - Stream
    - Clg official link

📚 6. Resource Management

- Add Learning Resource
- Upload PDF
- Edit Resource
- Delete Resource

🗺️ 7. Career Roadmap Management

- Add Roadmap Steps
- Edit Roadmap
- Delete Roadmap

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8. Expert Management

- Add Expert
- Edit Expert
- Delete Expert
- Assign Specialization

👥 9. Student Management

- View Students
- View Profiles
- Deactivate User

📈 10. Analytics & Reports

- Career Popularity Chart
- Stream Distribution Chart
- District Wise Usage
- Assessment Attempts

✅ **AI / ML FEATURES**

- Random Forest Classifier
- Career Prediction
- Probability-Based Output
- Feature Scaling
- Label Encoding
- Top-3 Prediction Selection
- Domain Filtering Rules
- Model Persistence (Saved Model)